

EXPERIMENT 4

Data Modelling and Form Design for a Business Application in Zoho Creator

Library Book Issue Request Form — Basic Fields Use Case

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Aim

To design a data model and build a business application form in Zoho Creator for managing library book issue requests, and to demonstrate the creation of form fields, the linking of related forms, and the generation of report data from submitted records.

Procedure

1. Log in to Zoho Creator and create a new application titled “Library Book Issue Management”.
2. Create the first form, “Member Profile”, and add fields to capture member details such as Member Name, Register/Roll Number, Member Type, Email Address, Contact Number, and Hostel/Residential Address.
3. Create the second form, “Book Issue Request”, and establish a lookup relationship with the Member Profile form using the Member field so that member records can be linked to each book request.
4. Add the remaining fields to the Book Issue Request form, including Book Category, Subject Areas of Interest, Number of Copies, Purpose of Borrowing, Preferred Issue Date, Preferred Collection Time, Agree to Library Rules, Request Status, and Processed By.
5. Configure field properties in the form builder, such as field link names, choice values for radio/checkbox fields, mandatory validation, and data privacy (PII) settings.
6. Preview and fill out the Book_Issue_Request_Form with sample member and request data, and submit the form.
7. View the submitted record in the Book_Issue_Request_Form Report to verify that the data has been captured and stored correctly.

Data Model

The application uses a set of related forms modelling a library management system. Member records are linked to Loan records, which in turn reference Book_Copy and Book records; Book records are further linked to Author, Publisher, and Category. Reservation and Fine_Payment forms extend the model to handle book reservations and fine collection. These lookup relationships allow a single member to raise multiple requests while keeping master data centralised and consistent.

LIBRARY MANAGEMENT SYSTEM – DATA MODEL

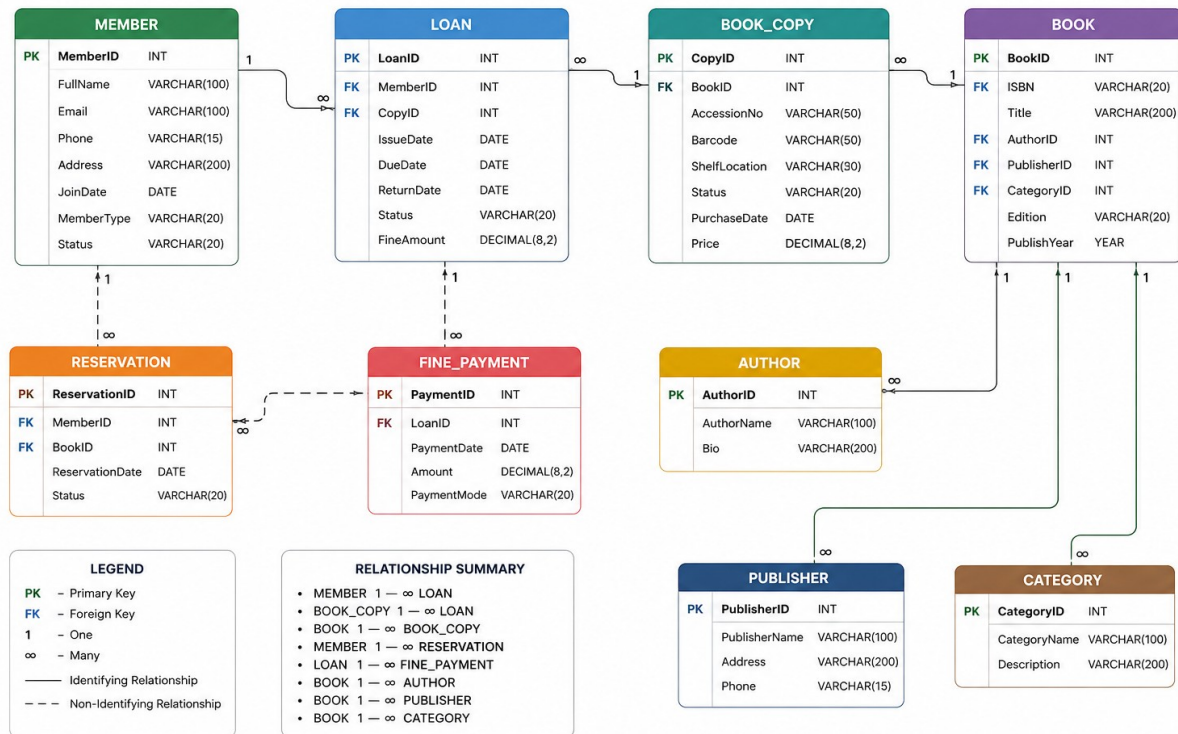


Fig. 1 — Data model showing the related library management system forms and their lookup relationships

Form	Key Fields	Purpose
Member Profile	Member Name, Register/Roll Number, Member Type, Email Address, Contact Number, Hostel/Residential Address	Stores master data for each library member
Book Issue Request	Member (lookup), Book Category, Subject Areas Of Interest, Number Of Copies, Purpose Of Borrowing, Preferred Issue Date, Preferred Collection Time, Agree To Library Rules, Request Status, Processed By	Captures each book request linked to a member

Explanation of Screenshots

The following screenshots illustrate the complete workflow: filling out the live library form as seen by a user, and the output data captured after submission.

1. The Form

The screenshot below shows the live form being filled out with sample data, including Name (First/Last), Phone number, a hand-drawn Librarian Signature field, Book Format (Hardcover), Fine Amount, Book Identifier, Book Website URL, and Reader Reviews link fields, before the record is submitted using the submit button.

The screenshot shows a web application titled "library management system". On the left is a dark sidebar with a logo and menu items: "Management Systems" and "library management system". The main area contains a form with the following fields:

- Name:** First Name "Ishith", Last Name "c".
- Phone:** "+91 843083089".
- Librarian Signature:** A drawing of a signature with a "[Clear]" link.
- Book Format:** "Hardcover" (dropdown menu).
- Fine Amount:** "500" (input field).
- Book Identifier:** "45343" (input field).
- Book Website URL:** "www.maasskdt.com" (input field).
- Link Name:** (empty input field).
- Reader Reviews:** "https://" (input field).

At the bottom of the form is a blue button labeled "Library". The top right of the application shows "Trial expires in 15 days" and links for "Upgrade", "Edit this application", and "Help". The bottom left of the sidebar shows a user profile for "AJAY K 2414...".

Fig. 2 — Library management system form filled out with sample member and book data

2. Output Data

After the form is submitted, the record appears in the All Management Systems report, confirming that the data has been stored correctly in the application. The report below displays the submitted record for “AJAY 241401006”, including phone number, librarian signature, book format (Hardcover), and fine amount (₹500.00).

The screenshot shows the "All Management Systems" report. It features a table with the following columns: Name, Phone, Librarian Signature, Book Format, Fine Amount, Book Identifier, and Book Website URL. The table contains two rows of data:

Name	Phone	Librarian Signature	Book Format	Fine Amount	Book Identifier	Book Website URL
AJAY 241401006	+919360774210		Hardcover	₹ 500.00		
aja						

Below the table, it says "Showing 2 of 2". The sidebar on the left is the same as in Fig. 2, but the "All Management Systems" option is highlighted. The top right of the application shows "Trial expires in 15 days" and links for "Upgrade", "Edit this application", and "Help".

Fig. 3 — All Management Systems report showing the submitted record

Result

A data model with related forms for a library management system was successfully designed in Zoho Creator. The form was built with appropriate field types and validations, tested by submitting sample data, and the submitted record was verified in the corresponding report, thereby demonstrating a working business application.